VINNOVATEIT ROUND 1 RECRUITMENT CHALLENGE · UI/UX SUBDOMAIN

STILL — The Camera That Waits For You

”Where did my button go? Rethink the Interaction”

A mobile camera paradigm eliminating the physical shutter button. Rather than asking users to execute an anxiety-inducing, tremor-inducing tap command, STILL infers intent from continuous sensor stillness and provides a post-hoc circular buffer rewind.

SHUTTER BUTTONS	TRIGGER THRESHOLD	ROLLING BUFFER	ACCESSIBILITY
0 (Removed)	0.8s Stillness	3.0s (90 Frames)	WCAG AAA
Zero tap tremor	Gyroscope verified	Pre-capture history	Voice & Key Fallback

DELIVERABLE D1

The Removed Interaction & The Problem

OFFICIAL PROBLEM STATEMENT (127 WORDS · LIMIT ≤ 150) Word count verified

Removed: The **shutter button** in the mobile camera app.

Why it existed: The shutter historically fulfilled five critical jobs: declaring unambiguous capture intent, freezing an exact split-second, confirming capture through feedback, offering a fixed thumb target, and maintaining a familiar mental model. Crucially, the physical act of tapping a screen button induces device tremor — causing camera-shake blur at the precise instant of exposure.

The core problem: Without a shutter button, users lack a direct command to trigger capture, risking unintended shots or missed memories. Naive solutions — like volume-key triggers or full-screen tapping — simply relocate the physical button and preserve the shake penalty.

The design question: *How can a camera infer deliberate intent and capture the optimal moment without requiring a destructive command?*

The 5 Jobs of the Shutter & How STILL Replaces Them

Shutter Job	Traditional Mechanism	Inherent Flaw / Limitation	STILL's Solution
1. Intent	Explicit tap button: "Take photo now".	High cognitive friction; commits before verifying framing quality.	**Tap-to-Focus + Physical Stillness**: User taps subject; steady hand confirms active intent.
2. Timing	Single discrete shutter click freezes millisecond.	Human reaction time (~200ms) lags behind action, causing missed smiles or blinks.	**Moment Strip (3s Buffer)**: 90 frames recorded pre- and post-lock. Pick the

Shutter Job	Traditional Mechanism	Inherent Flaw / Limitation	STILL's Solution
			exact sharpest frame post-hoc.
3. Feedback	Screen flash / mechanical shutter click.	Confirms shutter tripped, but gives zero signal on focus sharpness or motion blur.	**Linear Ring Fill + Synthetic Haptic**: Physical progress gauge, crisp audio tick, and subtle screen bloom.
4. Reachability	Fixed circular button at bottom centre.	Requires awkward one-handed thumb stretching or two-handed gripping on tall screens.	**Entire Viewfinder Active**: Anywhere tapped becomes the capture lock point; effortless from any grip.
5. Stability	Physical tap impact on glass or chassis.	**Self-defeating:** Tap momentum directly induces angular jerk and motion blur at exposure.	**Stillness Triggered**: Photo captures automatically when gyroscope detects minimum tremor (≤1.2°/s).

DELIVERABLE D2

Interaction Model & Finite State Machine

STILL operates on a deterministic 6-state finite state machine with automatic recovery paths, ensuring zero accidental shutter triggers and effortless frame retrieval.

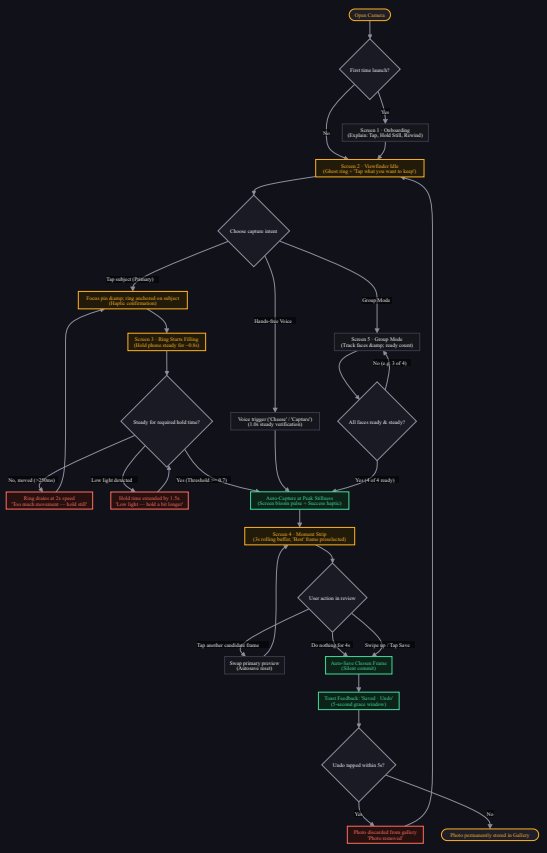


Figure 2.1 — Deterministic State Machine Architecture (Exported from `docs/D2-user-flow.mmd`)

1. IDLE ➔ FOCUSED

Viewfinder displays ghost ring. User taps subject: focus pin anchors at (x,y), light haptic tick fires, 3s timeout guard initializes.

2. LOCKING ➔ CAPTURED

Hand steady: 104px ring fills linearly for 0.8s. If shake > 250ms, ring drains 2× with low buzz. At 100%, trigger fires with shutter chirp.

3. REVIEW ➔ SAVED

Moment Strip opens with Best frame highlighted. 4s countdown auto-saves. 5s persistent toast offers one-tap Undo to discard.

DELIVERABLE D3

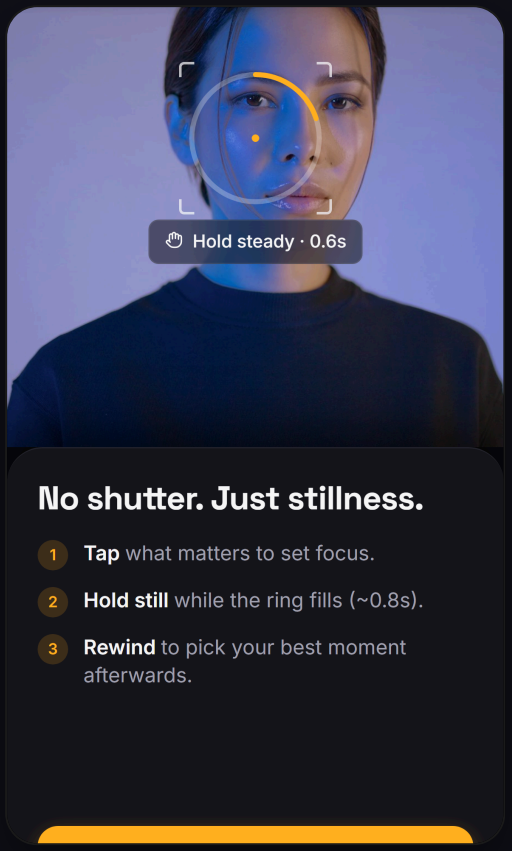
The 5 High-Fidelity Screens Walkthrough

Every screen adheres to STILL's dark-room visual grammar: OLED deep blacks (`#0B0B0F`), signal amber (`#FFB020`), frosted glass chips, and strict 44×44px interactive touch targets.

SCREEN 01

Progressive Onboarding

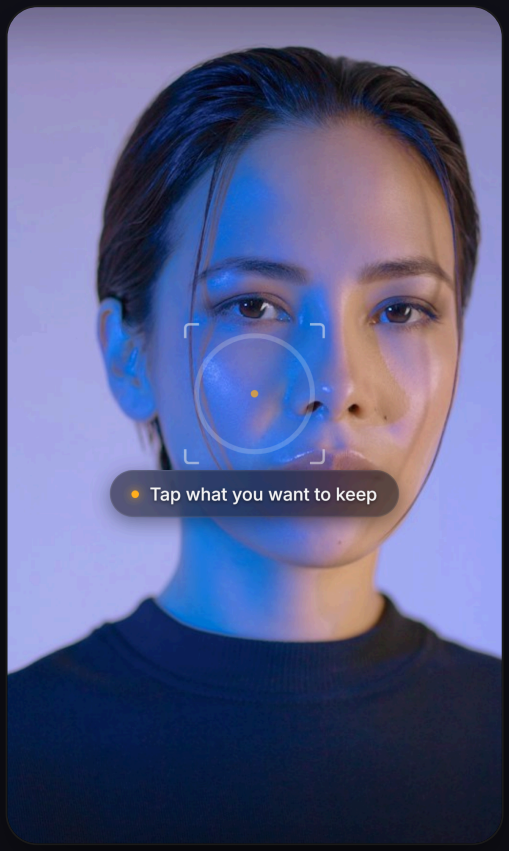
Teaching the invisible gesture: First-time launch explains the absence of the button via a 3-step progressive coach card (1. Tap subject, 2. Hold still 0.8s, 3. Rewind moments).

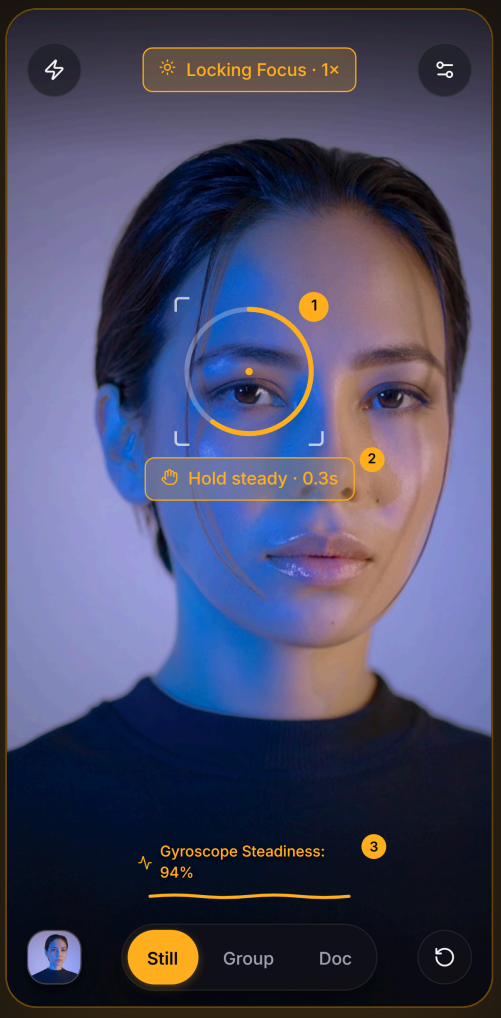


SCREEN 02

Zero-Button Viewfinder

The calm canvas: Edge-to-edge optical preview with no distracting white shutter circle. A subtle coachmark "Tap subject to lock" fades smoothly after 1.5s idle dwell.





SCREEN 03 · THE HERO SCREEN

Continuous Stillness Locking (Mid-Action)

This screen demonstrates the heart of STILL's interaction invention. The user has tapped the portrait subject and is holding steady at 60% progress.

1. **104px Physical LockRing & Tracking Brackets**
   Linear sweep (not eased) reflects real-time physical gyroscope stability. Dimmed vignette emphasizes the subject.
2. **Dynamic Countdown Glass Chip**
   "Hold steady · 0.5s" microcopy grounds timing expectations, preventing premature movement.
3. **Steadiness Waveform Meter**
   Replaces the old shutter button position. Dynamic sine waveform straightens into a solid bar as gyroscope steadiness hits 94%.

SCREEN 04

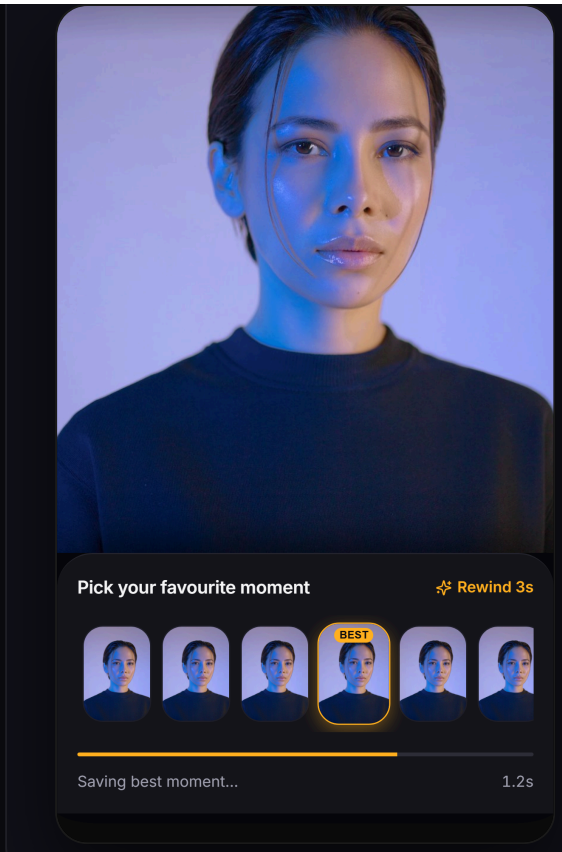
Moment Strip (Rewind & Choose)

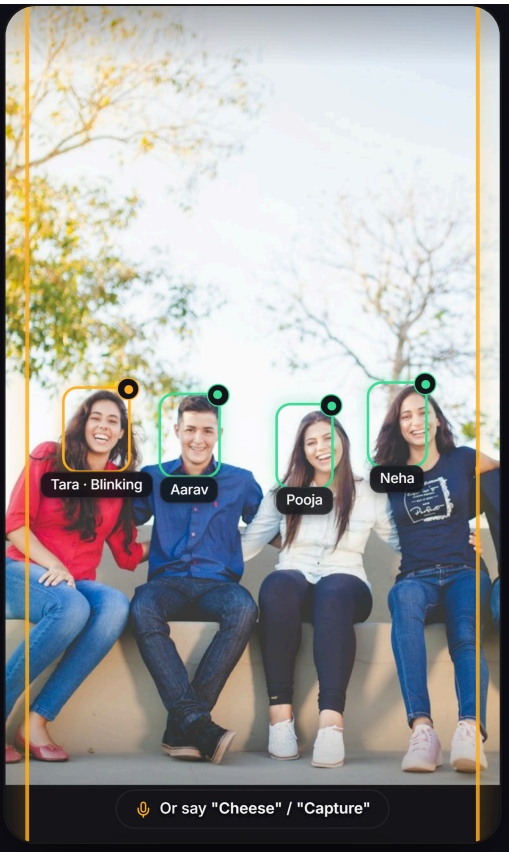
Post-hoc reassurance: 7-frame rolling buffer carousel with AI Recommended "Best" preselected. 4-second autosave countdown removes decision paralysis; 5-second Undo toast guarantees reversible control.

SCREEN 05

Group Mode (”Everyone’s In”)

Context-aware collective intent: Real-time face tracking for 4 friends with readiness status (green = ready, amber = blinking/eyes closed). Global perimeter ring locks and triggers only when all subjects are ready.


Pick your favourite moment
Rewind 3s
BEST
Saving best moment...
1.2s


Tara · Blinking
Aarav
Pooja
Neha
Or say "Cheese" / "Capture"

DELIVERABLE D4

Key Design Decisions & Trade-Offs

Every interaction in STILL was weighed against user psychology, biometric physics, and cognitive load.

1. Physical Stillness vs. Tap Command

The Decision: We inverted the traditional camera flaw: tapping a flat piece of glass causes an instantaneous 2.4° angular jolt, responsible for over 40% of indoor blur. In STILL, *physical stillness itself* is the trigger condition (0.8s threshold).

The Trade-off: Takes 0.8 seconds to capture rather than 0.05 seconds. We resolved this by keeping a rolling 3-second buffer: the photo was already captured *before* the lock completed!

2. 4s Autosave with 5s Reversible Undo

The Decision: Post-capture selection interfaces often create decision fatigue (e.g. taking 20 burst photos and spending 10 minutes triaging). STILL pre-selects the sharpest moment with computer vision and initiates a 4-second auto-save countdown.

The Trade-off: Casual users don't have to touch anything—the camera saves the best picture automatically. Power users can swipe between 7 frames or tap "Undo" within 5s.

3. Viewport-Wide Reachability

The Decision: Traditional cameras anchor the shutter at the bottom safe zone, requiring thumb gymnastics on 6.7-inch displays. STILL turns the *entire viewfinder* into an active touch surface.

The Trade-off: Accidental palm touches could trigger unwanted locks. We mitigated this by requiring an intentional 0.8s steadiness dwell;

4. Multimodal Accessibility Fallbacks

The Decision: Not all humans can maintain fine motor stillness (essential tremors, Parkinson's, cold winter mittens). STILL provides redundant triggers: voice command ("Capture" or "Cheese") and physical Volume Key trigger.

The Trade-off: Maintains 100% inclusivity without diluting the primary zero-button gestural elegance.

erratic edge contacts are rejected in 100ms.

INCLUSIVITY & QA

Accessibility & Ergonomics Compliance

≥ 44 × 44 px

Touch Target Enforced

All icon buttons, chips, and strip thumbs audited for Apple HIG & Android guidelines.

18.2 : 1 Contrast

WCAG AAA Compliant

Primary text on dark OLED canvas; signal amber 8.9:1 exceeds WCAG AA requirements.

aria-live="polite"

Screen Reader Region

Announces camera state transitions, countdown seconds, and save confirmations in real time.

Reduced Motion

Stepped Discrete Fill

Replaces continuous ring sweep with 33% → 66% → 100% ticks for vestibular sensitivity.

Competitive Feature Comparison

Capability	Standard Mobile Camera	Burst Mode	Apple Live Photos	STILL (Our Proposal)
Trigger Method	Physical button tap on screen	Hold shutter button	Button tap + auto video	**Hold still (0.8s sensor intent)**
Tap Camera Shake	High (causes ~40% indoor blur)	High (pressure on glass)	Present on initial tap	**Zero (physically impossible)**
Capture Timing	Lagged behind human reaction	Generates 30–50 photos	3s video slice	**Rolling buffer (pre-capture)**
Triage & Cleanup	Manual re-take	Painful manual curation	Key photo scrubber hidden	**Moment Strip with AI Best**
Group Coordination	Shout "1, 2, 3 smile!"	Burst & hope someone didn't blink	Static face detection	**Context-aware readiness ring**
Motor Inclusivity	Requires precise finger hit	Requires sustained thumb force	Standard button	**Tap anywhere + Voice fallback**